

MyDeliveryServiceApi Module

API contracts (interfaces, enums, model interfaces, validators, utilities) shared across service implementation, web service, and presentation layers.

Packages Overview

- `com.tnt.express.warp.mydelivery.service.api` : Core service interfaces & domain abstractions.
- `com.tnt.express.warp.mydelivery.model.api` : Data model interfaces consumed / produced by services.
- `com.tnt.express.mydelivery.validator` : Validation result & rule violation abstractions.
- `com.tnt.express.mydelivery.validator.annotation` : Custom validation annotations & validators.
- `com.tnt.express.mydelivery.service.email.api` : Email template data contracts.
- `com.tnt.express.mydelivery.util` : Shared utility helpers (sorting, country phone metadata).

Service API Interfaces

- `APCAc1` : Abstraction to APC (external) system operations.
- `Address` : Domain representation of an address (getter/setter contract).
- `ConsignmentDetails` : Contract exposing consignment-level shipment, status and redelivery metadata.
- `DeliverySystemParameter` : Represents a dynamic system parameter key/value pair.
- `DepotAvailability` : Abstraction of depot opening times / availability windows.
- `EmailSender` : Contract for sending templated emails (likely asynchronous / queued).
- `MyDeliveryPhoneNumber` : Structured phone number with type & validation semantics.
- `MyDeliveryPhoneNumberType` : Enum describing number categories (MOBILE, LANDLINE, etc.).
- `MyDeliveryService` : Primary façade for MyDelivery business operations (consignment lookup, options, confirmation, etc.).
- `MyDeliveryStatusCode` : Enum of internal workflow state / error or user journey codes.
- `PtlFormData` : Data needed to print a "Pending To Leave" (PTL) form.
- `RedeliveryConsignment` : Aggregate describing a consignment and permissible redelivery actions.
- `RedeliveryConsignmentParameters` : Mutable / builder style parameters for requesting redelivery.
- `RedeliveryOption` : Enum of supported redelivery modes (ORIGINAL_ADDRESS, ALTERNATIVE_ADDRESS, NEIGHBOUR, SELF_COLLECTION, LEAVE_IN_ABSENCE...).

- **RedeliveryRequest** : Immutable representation of a chosen redelivery request.
- **RedeliveryRequestBean** : Mutable form/backing bean for user input before confirmation.
- **RedeliveryRequestFactory** : Factory building validated **RedeliveryRequest** instances from beans / parameters.
- **ReportConstants** : Constants used for report parameter names, template ids, resource keys.
- **ReportPrinter** : Generic report generation contract (JasperReports integration expected).
- **ScreenNavigationStep** : Enum describing UI flow screens.
- **SelfCollectionDepotInformation** : Aggregate for depots supporting self collection (list, geocodes, opening hours).
- **StreamServeAcl** : Abstraction for calling StreamServe document generation service.
- **SupplyPostcodeRule** : Rule enumeration for postcode supply/validation logic.

Model Interfaces

- **AvailableOptionsModel** : Encapsulates the set of available redelivery options and any violations/warnings.
- **BaseResponseModel** : Common response structure (success flags, violations, timestamps).
- **ConfirmationModel** : Data returned after a successful confirmation (references, print tokens, etc.).
- **CountryMetadataModel** : Country-specific rules (postcode formats, phone prefixes, locale config).
- **DatesModel** : Collection of available delivery date slots (with logic for ordering / earliest date).
- **DepotAddressModel** : Basic depot address details.
- **DepotDateAndTimesModel** : Per-depot date/time window (opening hours, earliest collection date).
- **PrintConfirmationModel** : Data for printing confirmation receipt.
- **PrintPTLModel** : Data for printing PTL forms (date formatting, barcode id, etc.).
- **SelfCollectionModel** : Data shown during self collection option (depots, dates, selection state).
- **ViolationsModel** : Aggregated validation / rule violations (codes, severities, messages).

Validator Core

- **ErrorContainer** : Holds validation errors grouped by field or category.
- **FieldLengthConstants** : Centralized field max/min length constraints.
- **IEnumErrorCode** : Interface for error enums (code -> default message mapping).
- **MyDeliveryRuleViolation** : Represents a single violated rule (code, severity, field, message).
- **Severity** : Enum for violation severity (INFO/WARN/ERROR/FATAL etc.).
- **ValidationResult** : Interface conveying success state and list of violations.
- **ValidationResultImpl** : Concrete implementation of **ValidationResult**.

- `ValidatorFactory` : Produces validators for beans / requests (likely OVal or custom logic integration).

Validation Annotations

- `annotation.NotNullOrEmpty` : Custom annotation enforcing non-null/non-empty string.
- `annotation.IsValidEmail` : Annotation for email format validation.
- `annotation.impl.NotNullOrEmptyValidator` : Logic backing `NotNullOrEmpty`.
- `annotation.impl.IsValidEmailValidator` : Email validation logic (regex + potential domain rules).

Email Data Contracts

- `RedeliveryAlternativeDeliveryAddressData`
- `RedeliveryCollectFromTNTData`
- `RedeliveryLeaveInMyAbsenceAtNeighbourData`
- `RedeliveryLeaveInMyAbsenceWithInstructionsData`
- `RedeliveryOriginalAddressData` Each provides strongly typed getters exposing all tokens required to render its corresponding email template variant.

Utilities

- `CountryPhonePrefixes` : Mapping of country -> phone prefix list (used for phone validation & UI drop-downs).
- `SortByValueComparator` : Generic comparator to sort map entries by value.
- `Util` : Miscellaneous helper methods (string trimming, safe parsing, etc.).

Model API Usage Patterns

These interfaces are implemented in the `MyDeliveryServiceImpl` module under `model.impl`. They are serialized back to clients via web service endpoints (`MyDeliveryWebService`), or adapted into presentation beans in the presentation module.

Extension Points

- New redelivery option: add enum, extend factory, update `AvailableOptionsModel` & UI decorators.
- New validation rule: implement `IEnumErrorCode` constant, integrate into validation helper/factory.
- New email template: create API data interface and implementation + email sender adaptation.